



# ===== MASTER TUTOR SCRIPT: MHF4U UNIT 2 - PRACTICE MASTERY Student: Rigden Total Slides: 40

HOW TO USE THIS SCRIPT: Keep this text file open on a second monitor. The presentation file itself contains zero teacher notes to ensure it is 100% student-safe. Guide Rigden through the interactive tabs (Think, Hint, Solution) on each slide using these prompts.

SLIDE 1: Title Slide [SAY] Welcome back, Rigden! You know the theory inside and out. Today is a pure problem-solving masterclass. [SAY] We have 38 distinct challenges ahead of us, ranging from warm-ups to ultimate boss fights. [ASK] Do you have your paper, pencil, and calculator ready to go?

## LEVEL 1: WARM-UP (Slides 2 - 6)

SLIDE 2: Q1 - The Efficiency Test (Remainder Theorem) [ASK] Before you calculate, what is the fastest way to find a remainder? Long division, synthetic, or a theorem? [EXPECT] The Remainder Theorem. Plug in  $x = 2$ . [SAY] Exactly. Go ahead and evaluate  $P(2)$ .

SLIDE 3: Q2 - Factor Theorem Basics [ASK] How do we definitively prove if  $(x+3)$  is a factor without dividing? [EXPECT] Plug in  $-3$ . If it equals 0, it's a factor. [SAY] Let's test it. What is  $P(-3)$ ?

SLIDE 4: Q4 - Missing Placeholders [ASK] Look at the polynomial  $x^3 - 8$ . If you set up synthetic division, what numbers go in your top row? [EXPECT] 1, 0, 0,  $-8$ . [TIP] Emphasize that forgetting zeros is the #1 reason students fail division questions.

SLIDE 5: Q5 - Reading Synthetic Division [ASK] The bottom row is 2,  $-1$ , 4, | 0. If we started with a cubic ( $x^3$ ), what degree does the quotient start with? [EXPECT] Squared ( $x^2$ ). So the quotient is  $2x^2 - x + 4$ .

SLIDE 6: Q6 - End Behaviour Check [ASK] Just a quick Unit 1 check. Degree is 4, leading coefficient is negative. Where does it point? [EXPECT] Q3 to Q4. Both down.

## LEVEL 2: MODERATE (Slides 7 - 14)

SLIDE 7: Q7 - Working Backwards for 'k' [ASK] We know the remainder is 7 when divided by  $(x+1)$ . How do we set up the algebra? [EXPECT] Set  $P(-1) = 7$ . [SAY] Perfect. Substitute  $-1$ , set the whole thing to 7, and isolate  $k$ .

SLIDE 8: Q8 - Find 'k' with a Factor [ASK] This is identical to the last question, but it says  $(x-2)$  is a factor. What is the hidden remainder? [EXPECT] The remainder is 0. So  $P(2) = 0$ .

SLIDE 9: Q9 - The  $(ax-b)$  Trap [ASK] We are dividing by  $(2x - 1)$ . What root goes outside the synthetic bracket? [EXPECT]  $1/2$ . [TIP] Remind him: After doing synthetic division with a fraction, you MUST divide the quotient coefficients by the denominator (2).

SLIDE 10: Q10 - Division Statement [SAY] A division statement is just Dividend = Divisor \* Quotient + Remainder. [ASK] Can you piece those four parts together for me?

SLIDE 11: Q11 - Factoring a Cubic (Step 1) [ASK] We have  $x^3 - 4x^2 + x + 6$ . What numbers should we guess and check to find the first factor? [EXPECT] Factors of 6: 1,  $-1$ , 2,  $-2$ , 3,  $-3$ , 6,  $-6$ . [SAY] Let's try  $P(-1)$  first.

SLIDE 12: Q12 - Factoring a Cubic (Step 2) [SAY] Now that you know  $(x+1)$  is a factor, use synthetic division to break it down. [ASK] What quadratic are you left with?

SLIDE 13: Q13 - Sum of Cubes [SAY] This is a special Grade 12 formula.  $a^3 + b^3$ . Do you remember the SOAP acronym for the signs? [EXPECT] Same, Opposite, Always Positive. [SAY] Yes!  $(a+b)(a^2 - ab + b^2)$ .

SLIDE 14: Q14 - Difference of Cubes [SAY] Apply the same SOAP rule to  $8x^3 - 27$ . What are your 'a' and 'b' values? [EXPECT]  $a = 2x$ ,  $b = 3$ .

### LEVEL 3: ADVANCED (Slides 15 - 24)

SLIDE 15: Q15 - The Systems Challenge (a and b) [ASK] We have two unknowns, 'a' and 'b'. What mathematical concept do we need to solve this? [EXPECT] A system of linear equations. Two equations! [SAY] Use the two remainders given to build your two equations, then use elimination.

SLIDE 16: Q16 - Systems Challenge Part 2 [SAY] Continue from the last slide. Substitute your 'a' value back in to find 'b'.

SLIDE 17: Q17 - Factoring a Quartic (Degree 4) [ASK] We have a degree 4 polynomial. How many times will we have to do synthetic division to get it down to a quadratic? [EXPECT] Twice. [SAY] Exactly. Find one factor, divide. Find a second factor of the NEW quotient, divide again.

SLIDE 18: Q18 - Disguised Quadratic [ASK] Look at  $x^4 - 5x^2 + 4$ . It's a quartic, but does it look like something easier? [EXPECT] It looks like a quadratic trinomial. [SAY] Yes! Let  $m = x^2$ . Factor it normally, then substitute  $x^2$  back in.

SLIDE 19: Q19 - Factor by Grouping [SAY] Sometimes we can skip synthetic division completely. Look at  $x^3 + 2x^2 - 9x - 18$ . [ASK] Can you common factor the first two terms, and the last two terms separately?

SLIDE 20: Q20 - Multiplicity Graphing [ASK] You have the factored form. Tell me what happens at  $x = -2$  (order 2) and  $x = 1$  (order 3). [EXPECT] Bounces at -2, inflects at 1.

SLIDE 21: Q21 - Solving Polynomial Equations [SAY] Solving  $P(x) = 0$  is the exact same as finding x-intercepts. Bring everything to one side and factor completely.

SLIDE 22: Q22 - Irrational Roots [SAY] You did the synthetic division, but the remaining quadratic doesn't factor perfectly. [ASK] What Grade 10 formula do we use when a quadratic won't factor? [EXPECT] The Quadratic Formula.

SLIDE 23: Q23 - Complex Systems [SAY] Another 'a' and 'b' system, but this time one of the conditions is a "factor". What is the remainder when it's a factor? (Zero!).

SLIDE 24: Q24 - Sketching from Standard Form [SAY] Walk me through the entire 4-step process for graphing this standard form cubic. [EXPECT] 1. Factor theorem for 1st root. 2. Synthetic div. 3. Factor quadratic. 4. Plot intercepts and end behaviour.

### LEVEL 4: CHALLENGE (Slides 25 - 32)

SLIDE 25: Q25 - Families of Polynomials [ASK] We know the x-intercepts, so we can build the skeleton.  $y = a(x-r_1)(x-r_2)$ . How do we find 'a'? [EXPECT] Plug in the given point (x, y) and solve for 'a'.

SLIDE 26: Q26 - Families (Tangent/Bounce) [SAY] The problem says the graph is "tangent" to the x-axis at  $x = 3$ . What does that mean for the exponent? [EXPECT] Tangent means bounce, so the exponent is 2.

SLIDE 27: Q27 - Polynomial Inequalities (Visual) [SAY] The question asks where  $P(x) > 0$ . That simply means: Where is the graph above the x-axis? [ASK] Looking at the roots, what intervals is the graph physically above the line?

SLIDE 28: Q28 - Polynomial Inequalities (Algebraic) [SAY] To solve  $x^2 - x - 6 < 0$ , first find the roots. Then, sketch a quick mini-graph to see where the parabola dips below zero.

SLIDE 29: Q29 - Word Problem (Volume of a Box) [SAY] Volume = Length \* Width \* Height. The volume is given as a cubic polynomial. [ASK] How do we find the dimensions (L, W, H)? [EXPECT] We have to factor the cubic into three linear brackets!

SLIDE 30: Q30 - Error Analysis (Synthetic Setup) [ASK] Look at the student's work. They divided by  $(x+4)$ . What mistake did they make on the outside of the bracket? [EXPECT] They put +4 instead of -4.

SLIDE 31: Q31 - Error Analysis (The ax-b trap) [ASK] The student did synthetic division for  $(3x-1)$  and got a remainder of 0. But their factored form is wrong. Why? [EXPECT] They forgot to divide the quotient by 3!

SLIDE 32: Q32 - Graph to Equation (Degree 4) [ASK] Look at the graph. It looks like a 'W'. It bounces at -2 and crosses at 1 and 3. Build the skeleton!

## LEVEL 5: MASTER CHALLENGE (Slides 33 - 38)

SLIDE 33: Q33 - Factoring Degree 5 [SAY] This is a massive degree 5 polynomial. But look closely—can you common factor an 'x' out first? [EXPECT] Yes. Then it becomes a quartic, and we use synthetic division twice.

SLIDE 34: Q34 - Complex Word Problem [SAY] The height of a roller coaster is given by  $h(t)$ . When is the coaster at ground level? [EXPECT] Set  $h(t) = 0$  and factor. Time (t) must be positive.

SLIDE 35: Q35 - Undetermined Coefficients [SAY] Expand both sides algebraically and compare the coefficients of the  $x^2$  terms and x terms to find 'a' and 'b'.

SLIDE 36: Q36 - Remainder Theorem Trap [SAY]  $P(x)$  is divided by  $x^2 - 4$ . Since the divisor is degree 2, the remainder must be degree 1 ( $ax + b$ ). [TIP] This is a highly advanced contest-style question. Guide him to set up  $P(2)$  and  $P(-2)$  to create a system.

SLIDE 37: Q37 - The Inequality Sign Flip [ASK] When solving  $-2(x-3)(x+1) > 0$  algebraically, what happens if you divide both sides by -2? [EXPECT] You MUST flip the inequality sign!

SLIDE 38: Q38 - The Ultimate Boss [SAY] This question combines systems of equations, the factor theorem, quadratic formula, and sketching. Take your time. Walk me through your entire battle plan.

SLIDE 39: Key Takeaways [SAY] Take a screenshot of this. These are the core rules that will save you on the unit test.

SLIDE 40: Homework & Sign Off [SAY] Incredible endurance today, Rigden. You survived 38 intense problems. Tackle the remaining Jensen math reviews for homework. See you next time!